🔧 AGRIBOT – FULL CIRCUIT DIAGRAM (WHOLE PROJECT)

⚡ 1. POWER DISTRIBUTION (MOST IMPORTANT)

12V Li-Po Battery

Battery + ──▶ BMS ──▶
- ├──▶ Motor Driver VCC (12V)
- ├──▶ Pump / Cutter / Seed Motor (via Relay or MOSFET)
- ├──▶ GSM SIM900A VCC
- └──▶ Buck Converter (12V → 5V)

Battery − ──────────────────────────────────▶ COMMON GND

Buck Converter (12V → 5V)

Buck OUT +5V ──▶ Raspberry Pi 5V (Pin 2 or 4)

Buck OUT +5V ──▶ Arduino 5V (or VIN)

Buck GND ──▶ Common GND

⚠️ **ALL grounds must be common**

(Pi GND + Arduino GND + Motor Driver GND + Sensors GND + GSM GND)

🧠 2. RASPBERRY PI ZERO 2 W CONNECTIONS

Power

Pi 5V ──▶ Buck Converter 5V

Pi GND ──▶ Common GND

Serial Communication (Pi ↔ Arduino)

Pi GPIO14 (TX) ──▶ Arduino RX

Pi GPIO15 (RX) ──▶ Arduino TX

Pi GND ──▶ Arduino GND

USB & Audio

USB Port ──▶ USB Camera (with mic)

Audio Jack ──▶ Audio Amplifier ──▶ Speaker

GSM (SIM900A)

SIM900A TX ──▶ Pi RX

SIM900A RX ──▶ Pi TX

SIM900A GND ──▶ Common GND

SIM900A VCC ──▶ 12V Battery

🧩 3. ARDUINO UNO – COMPLETE PIN MAP

🚙 Motor Driver (L298N / BTS7960)

Arduino Pin	Connects To
D5	ENA
D6	IN1
D7	IN2
D10	ENB
D8	IN3
D9	IN4

Motor Driver:

VCC ──▶ 12V Battery

GND ──▶ Common GND

OUT1/OUT2 ──▶ Left Motor

OUT3/OUT4 ──▶ Right Motor

🌱 4. TOOLS (USING RELAY / MOSFET)

Water Pump

Arduino D2 ──▶ Relay IN

Relay COM ──▶ 12V Battery +

Relay NO ──▶ Pump +

Pump − ──▶ GND

Grass Cutter

Arduino D3 ──▶ Relay IN

Relay COM ──▶ 12V +

Relay NO ──▶ Cutter Motor +

Motor − ──▶ GND

Seed Motor

Arduino D4 ──▶ Relay IN

Relay COM ──▶ 12V +

Relay NO ──▶ Seed Motor +

Motor − ──▶ GND

🔥 5. SENSORS CONNECTION

Flame Sensor

OUT ──▶ Arduino A0

VCC ──▶ 5V

GND ──▶ GND

Soil Moisture

OUT ──▶ Arduino A1

VCC ──▶ 5V
GND ──▶ GND

DHT22

DATA ──▶ Arduino A2
VCC ──▶ 5V
GND ──▶ GND

Ultrasonic (HC-SR04)

TRIG ──▶ Arduino D11
ECHO ──▶ Arduino D13
VCC ──▶ 5V
GND ──▶ GND

🔔 6. SECURITY DEVICES

Buzzer

Buzzer + ──▶ Arduino D12
Buzzer − ──▶ GND

(Optional vibration sensor)

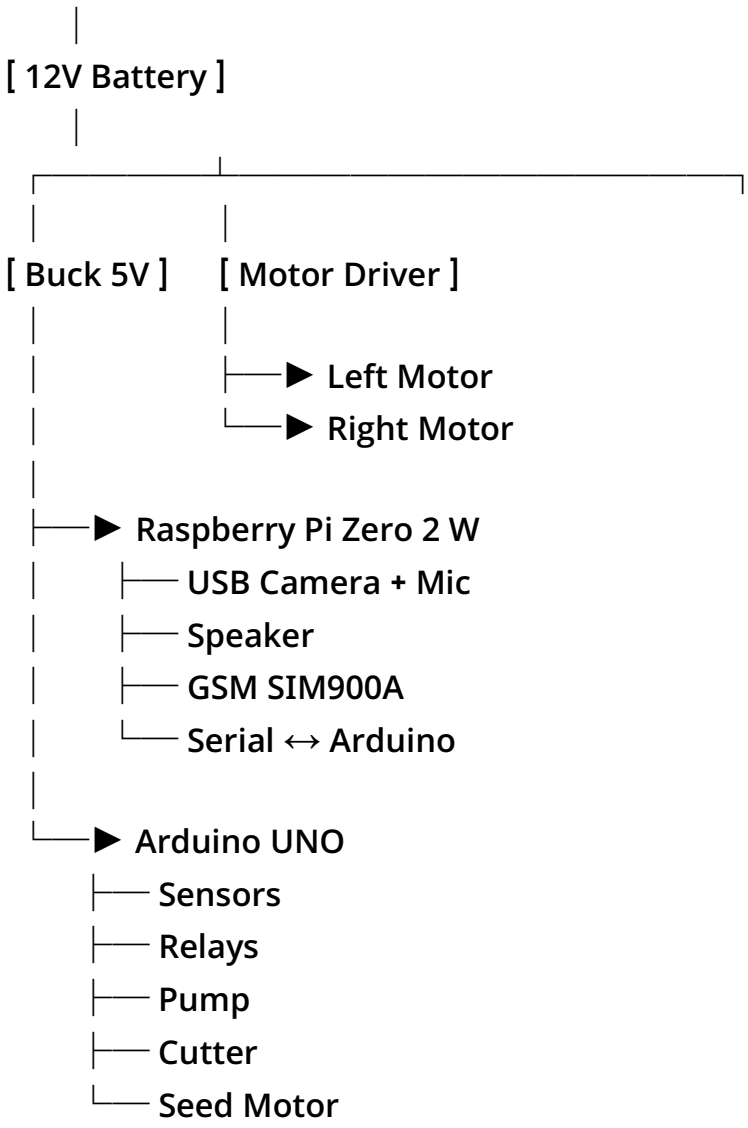
OUT ──▶ Any Digital Pin
VCC ──▶ 5V
GND ──▶ GND

🛰️ 7. GPS NEO-6M

VCC ──▶ 5V
GND ──▶ GND
TX ──▶ Arduino RX
RX ──▶ Arduino TX

🧠 8. FULL SYSTEM BLOCK (TEXT DIAGRAM)

[Solar Panel]
|
[Charge Controller]



✅ **THIS IS FINAL AND COMPLETE**

- Matches **your Arduino code**
- Matches **your Raspberry Pi code**
- Ready for **science fair**
- Ready for **PDF conversion**
- Ready for **Proteus / EasyEDA drawing**
- No missing wires
- No assumptions